

PRACTICAL-2**1 – CREATE TABLE**

```
mysql> USE bankdb;
Database changed
mysql> CREATE TABLE student_marks1 (
->     roll_no INT PRIMARY KEY,
->     name VARCHAR(50),
->     subject VARCHAR(50),
->     marks DECIMAL(5,2)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql> DESC student_marks1;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| roll_no | int           | NO   | PRI | NULL    |       |
| name    | varchar(50)   | YES  |     | NULL    |       |
| subject | varchar(50)   | YES  |     | NULL    |       |
| marks   | decimal(5,2)  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.01 sec)
```

2 – INSERT DATA

```
mysql> INSERT INTO student_marks1
-> (roll_no, name, subject, marks)
-> VALUES
-> (1, 'Ravi', 'Math', 85.50),
-> (2, 'Sita', 'Math', 92.75),
-> (3, 'Anil', 'Math', 78.40),
-> (4, 'Priya', 'Math', 88.90),
-> (5, 'Vijay', 'Math', 80.25),
-> (6, 'subbusir', 'aws', 98.50),
-> (7, 'dwaraka', 'dbms', 95.75),
-> (8, 'ranjani', 'english', 97.40),
-> (9, 'kavithamam', 'aws1', 99.90),
-> (10, 'seetha', 'azure', 82.25);
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM student_marks1;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1       | Ravi      | Math    | 85.50 |
| 2       | Sita      | Math    | 92.75 |
| 3       | Anil      | Math    | 78.40 |
| 4       | Priya     | Math    | 88.90 |
| 5       | Vijay     | Math    | 80.25 |
| 6       | subbusir  | aws     | 98.50 |
| 7       | dwaraka   | dbms    | 95.75 |
| 8       | ranjani   | english | 97.40 |
| 9       | kavithamam | aws1    | 99.90 |
| 10      | seetha    | azure   | 82.25 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

3 – SIMPLE SELECT

```
mysql> SELECT * FROM student_marks1;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
2	Sita	Math	92.75
3	Anil	Math	78.40
4	Priya	Math	88.90
5	Vijay	Math	80.25
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90
10	seetha	azure	82.25

```
10 rows in set (0.00 sec)
```

4 – Aggregate Functions

4(a) COUNT()

```
mysql> SELECT COUNT(*) AS total_students
-> FROM student_marks1;
```

total_students
10

```
1 row in set (0.02 sec)
```

4(b) SUM()

```
mysql> SELECT SUM(marks)
-> FROM student_marks1;
```

SUM(marks)
899.60

```
1 row in set (0.00 sec)
```

4(c) AVG()

```
mysql> SELECT AVG(marks)
-> FROM student_marks1;
+-----+
| AVG(marks) |
+-----+
| 89.960000 |
+-----+
1 row in set (0.00 sec)
```

4(d) MAX()

```
mysql> SELECT MAX(marks)
-> FROM student_marks1;
+-----+
| MAX(marks) |
+-----+
| 99.90 |
+-----+
1 row in set (0.00 sec)
```

4(e) MIN()

```
mysql> SELECT MIN(marks)
-> FROM student_marks1;
+-----+
| MIN(marks) |
+-----+
| 78.40 |
+-----+
1 row in set (0.00 sec)
```

5 – WHERE Clause**5(a) Marks Greater than 85**

```
mysql> SELECT * FROM student_marks1
-> WHERE marks > 85;
+-----+-----+-----+-----+
| roll_no | name      | subject | marks |
+-----+-----+-----+-----+
| 1 | Ravi      | Math    | 85.50 |
| 2 | Sita      | Math    | 92.75 |
| 4 | Priya     | Math    | 88.90 |
| 6 | subbusir  | aws     | 98.50 |
| 7 | dwaraka   | dbms    | 95.75 |
| 8 | ranjani   | english | 97.40 |
| 9 | kavithamam | aws1    | 99.90 |
+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

5(b) Marks greater than or equal to 90

```
mysql> SELECT * FROM student_marks1
-> WHERE marks >= 90;
```

roll_no	name	subject	marks
2	Sita	Math	92.75
6	subbusir	aws	98.50
7	dwaraka	dbms	95.75
8	ranjani	english	97.40
9	kavithamam	aws1	99.90

```
5 rows in set (0.00 sec)
```

5(c) Marks less than 80

```
mysql> SELECT * FROM student_marks1
-> WHERE marks < 80;
```

roll_no	name	subject	marks
3	Anil	Math	78.40

```
1 row in set (0.00 sec)
```

5(d) Marks between 80 and 90

```
mysql> SELECT * FROM student_marks1
-> WHERE marks BETWEEN 80 AND 90;
```

roll_no	name	subject	marks
1	Ravi	Math	85.50
4	Priya	Math	88.90
5	Vijay	Math	80.25
10	seetha	azure	82.25

```
4 rows in set (0.00 sec)
```

5(e) Name starts with 'P'

```
mysql> SELECT * FROM student_marks1
-> WHERE name LIKE 'P%';
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
|        4 | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

5(f) Name is Ravi, Sita, or Vijay

```
mysql> SELECT * FROM student_marks1
-> WHERE name IN ('Ravi','Sita','Vijay');
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
|        1 | Ravi  | Math    | 85.50 |
|        2 | Sita  | Math    | 92.75 |
|        5 | Vijay | Math    | 80.25 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

5(g) Marks > 85 and subject is Math or name starts with P

```
mysql> SELECT * FROM student_marks1
-> WHERE marks > 85
-> AND (subject='Math' OR name LIKE 'P%');
+-----+-----+-----+-----+
| roll_no | name  | subject | marks |
+-----+-----+-----+-----+
|        1 | Ravi  | Math    | 85.50 |
|        2 | Sita  | Math    | 92.75 |
|        4 | Priya | Math    | 88.90 |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

6 – UPDATE

6(a) Update the marks of roll number 3

```
mysql> UPDATE student_marks1
      -> SET marks = 90.00
      -> WHERE roll_no = 3;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM student_marks1
      -> WHERE roll_no = 3;
+-----+-----+-----+-----+
| roll_no | name | subject | marks |
+-----+-----+-----+-----+
|      3 | Anil | Math    | 90.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

6(b) Update the subject for students with marks less than 80

```
mysql> UPDATE student_marks1
      -> SET subject = 'Remedial Math'
      -> WHERE marks < 80;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

mysql> SELECT * FROM student_marks1
      -> WHERE subject = 'Remedial Math';
Empty set (0.00 sec)
```

7 – DELETE

7(a) Delete the record with roll number 5

```
mysql> DELETE FROM student_marks1
-> WHERE roll_no = 5;
Query OK, 1 row affected (0.01 sec)

mysql> ^C
mysql> SELECT * FROM student_marks1
-> WHERE roll_no = 5;
Empty set (0.00 sec)
```

7(b) Delete students with marks less than 75

```
mysql> DELETE FROM student_marks1
-> WHERE marks < 75;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM student_marks1
-> WHERE marks < 75;
Empty set (0.00 sec)
```

8 – ORDER BY

8(a) – Sort by Marks (Ascending)

```
mysql> SELECT * FROM student_marks1
-> ORDER BY marks ASC;
+-----+-----+-----+-----+
| roll_no | name   | subject | marks |
+-----+-----+-----+-----+
| 10      | seetha | azure   | 82.25 |
| 1       | Ravi   | Math    | 85.50 |
| 4       | Priya  | Math    | 88.90 |
| 3       | Anil   | Math    | 90.00 |
| 2       | Sita   | Math    | 92.75 |
| 7       | dwaraka | dbms    | 95.75 |
| 8       | ranjani | english | 97.40 |
| 6       | subbusir | aws     | 98.50 |
| 9       | kavithamam | aws1  | 99.90 |
+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

8(b) – Sort by Marks (Descending)

```
mysql> SELECT * FROM student_marks1
-> ORDER BY marks DESC;
```

roll_no	name	subject	marks
9	kavithamam	aws1	99.90
6	subbusir	aws	98.50
8	ranjani	english	97.40
7	dwaraka	dbms	95.75
2	Sita	Math	92.75
3	Anil	Math	90.00
4	Priya	Math	88.90
1	Ravi	Math	85.50
10	seetha	azure	82.25

9 rows in set (0.00 sec)

8(c) – Sort by Name (Ascending)

```
mysql> SELECT * FROM student_marks1
-> ORDER BY name ASC;
```

roll_no	name	subject	marks
3	Anil	Math	90.00
7	dwaraka	dbms	95.75
9	kavithamam	aws1	99.90
4	Priya	Math	88.90
8	ranjani	english	97.40
1	Ravi	Math	85.50
10	seetha	azure	82.25
2	Sita	Math	92.75
6	subbusir	aws	98.50

9 rows in set (0.00 sec)

8(d) – Sort by Name (Descending)

```
mysql> SELECT * FROM student_marks1
-> ORDER BY name DESC;
```

roll_no	name	subject	marks
6	subbusir	aws	98.50
2	Sita	Math	92.75
10	seetha	azure	82.25
1	Ravi	Math	85.50
8	ranjani	english	97.40
4	Priya	Math	88.90
9	kavithamam	aws1	99.90
7	dwaraka	dbms	95.75
3	Anil	Math	90.00

9 rows in set (0.00 sec)

9 – GROUP BY**9(a) – GROUP BY with SUM(marks)**

```
mysql> SELECT subject, SUM(marks) AS total_marks
-> FROM student_marks1
-> GROUP BY subject;
```

subject	total_marks
Math	357.15
aws	98.50
dbms	95.75
english	97.40
aws1	99.90
azure	82.25

6 rows in set (0.00 sec)

9(b) – GROUP BY with AVG(marks)

```
mysql> SELECT subject, AVG(marks) AS avg_marks
-> FROM student_marks1
-> GROUP BY subject;
```

subject	avg_marks
Math	89.287500
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000
azure	82.250000

```
6 rows in set (0.01 sec)
```

9(c) – GROUP BY with COUNT(*)

```
mysql> SELECT subject, COUNT(*) AS num_students
-> FROM student_marks1
-> GROUP BY subject;
```

subject	num_students
Math	4
aws	1
dbms	1
english	1
aws1	1
azure	1

```
6 rows in set (0.00 sec)
```

10 – HAVING**10(a) Display subjects whose average marks are greater than 90**

```
mysql> SELECT subject, AVG(marks) AS avg_marks
-> FROM student_marks1
-> GROUP BY subject
-> HAVING AVG(marks) > 90;
```

subject	avg_marks
aws	98.500000
dbms	95.750000
english	97.400000
aws1	99.900000

4 rows in set (0.00 sec)

10(b) Display subjects having more than one student

```
mysql> SELECT subject, COUNT(*) AS num_students
-> FROM student_marks1
-> GROUP BY subject
-> HAVING COUNT(*) > 1;
```

subject	num_students
Math	4

1 row in set (0.00 sec)